

SHUBHAKARINI K

Namakkal, Tamil Nadu

 karthikeyanshubhakarini@gmail.com |  8754398300

GitHub: github.com/Shubha-Karthi | LinkedIn: linkedin.com/in/shubhakarini-karthikeyan-795606268

PROFESSIONAL SUMMARY

Entry-level Software Engineer with strong skills in Python programming, problem-solving, and application development. Experienced in building software tools including search engines, file management systems, code analysis applications, and productivity solutions. Passionate about backend development, automation, and scalable software systems.

TECHNICAL SKILLS

Languages: Python, C

Core Concepts: OOP, Problem Solving, File Handling, Data Structures

Libraries & Technologies: Pandas (Basics), NumPy (Basics), Matplotlib (Basics), IoT Systems

Tools & Platforms: VS Code, Arduino IDE, Tinkercad, Wokwi, GitHub

PROJECTS

STUDENT MANAGEMENT SYSTEM | Python

- Developed a console-based application to manage student records with add, update, delete, and search functionalities.
- Implemented file handling techniques for secure data storage and efficient record retrieval.
- Applied CRUD operations and structured program flow for effective data management.
- Improved backend logic and user interaction through modular application design.

CODE COMPLEXITY ANALYZER | Python

- Built a Python-based tool to analyze source code complexity using nested loop and conditional detection.
- Implemented logical flow analysis and string parsing for evaluating code structure and readability.
- Generated complexity reports to identify maintainability and optimization issues in programs.
- Applied file handling and automation techniques for source code processing.

MINI SEARCH ENGINE | Python

- Developed a lightweight search engine to retrieve keywords from multiple text files efficiently.
- Implemented keyword indexing and optimized search logic for faster result retrieval.
- Applied text processing and file handling concepts for structured data searching.
- Improved information retrieval efficiency through organized keyword matching techniques.

URL SHORTENER | Python

- Built a Python-based URL shortener to convert long URLs into compact and manageable links.
- Implemented URL mapping and retrieval functionality using unique identifier generation techniques.
- Applied string manipulation and backend processing concepts for efficient link management.
- Improved data organization and application workflow through structured URL handling logic.

DUPLICATE FILE DETECTOR | Python

- Developed a duplicate file detection system using hashing and file comparison techniques.
- Automated directory scanning and duplicate report generation for storage optimization.
- Implemented efficient file analysis methods to identify identical files accurately.
- Applied automation and file management concepts to improve data organization.

SMART NOTES ORGANIZER | Python

- Designed a notes management application with add, edit, delete, and keyword-based search functionalities.
- Implemented categorized note organization for efficient storage and quick retrieval of information.
- Applied file handling and data management techniques for structured note processing.
- Improved application workflow and user interaction through modular program design.

INTERNSHIP EXPERIENCE

IoT Intern – RedMind Technologies | Jun 2025 – Jul 2025

- Developed and tested an automated appliance monitoring system using sensors and relay modules.
- Worked with real-time sensor data processing, debugging, and embedded system testing.

Intern – Wearable MEMS Company | Jan 2026 – Apr 2026

- Assisted in AI- and sensor-based proposal development and technical documentation.
- Supported data analysis, reporting, and system evaluation activities.

EDUCATION

Bachelor of Engineering – Electrical and Electronics Engineering

Government College of Engineering, Srirangam | 2022 – 2026

CGPA: 7.82